	ST5-XL	LASER2	MiniLM-L12 [34]	MPNet[35]	LaBSE	SGPT-BLOOM-1.7B	SGPT-BLOOM-7.1B
Embedding classification performance on MASSIVE (FitzGerald et al., 2022) scored using accuracy							
Arabic (ar)	4.18	37.16	51.43	45.14	50.86	54.59	**59.25**
Bengali (bn)	2.60	42.51	48.79	35.34	58.22	57.76	**61.59**
English (en)	**72.09**	47.91	69.32	66.84	61.46	66.69	69.67
Spanish (es)	57.97	45.44	64.43	59.66	58.32	61.77	**66.35**
French (fr)	60.99	46.13	64.82	60.25	60.47	64.58	**66.95**
Hindi (hi)	3.02	40.20	62.77	58.37	59.40	60.74	**63.54**
Indonesian (id)	41.53	45.81	**65.43**	59.85	61.12	60.07	64.06
Kannada (kn)	2.79	4.32	50.63	40.98	**56.24**	48.56	53.54
Malayalam (ml)	2.98	41.33	54.34	42.41	57.91	55.10	**58.27**
Portuguese (pt)	57.95	48.55	64.89	61.27	60.16	62.52	**66.69**
Swahili (sw)	30.60	31.89	31.95	29.57	**51.62**	43.90	49.81
Tamil (ta)	1.79	29.63	50.17	36.77	55.04	52.66	**56.40**
Telugu (te)	2.26	36.03	52.82	40.72	**58.32**	49.32	54.71
Urdu (ur)	2.70	26.11	56.37	52.80	56.70	51.00	**56.75**
Vietnamese (vi)	21.47	44.33	59.68	56.61	56.67	59.85	**64.53**
Semantic textual similarity on STS22 (Madabushi et al., 2022) scored using spearman correlation of cosine similarities							
Arabic (ar)	29.60	42.57	52.19	46.20	57.67	48.64	**58.67**
English (en)	64.32	39.76	63.06	61.72	60.97	61.45	**66.13**
Spanish (es)	58.16	54.92	59.91	56.56	63.18	61.81	**65.41**
French (fr)	77.49	58.61	74.30	70.55	77.95	73.18	**80.38**
Chinese (zh)	33.55	49.41	61.75	58.75	63.02	58.53	**66.78**

Table 10: Performance of BLOOM models finetuned for sentence embeddings on classification and STS datasets from MTEB (Muennighoff et al., 2022b).

2018; Tenney et al., 2018; Belinkov and Glass, 2019; Teehan et al., 2022), although it comes with certain shortcomings (Belinkov, 2022). Examination of the LLM embeddings can help shed light on the generalizing abilities of the model apart from its training objective loss or downstream task evaluation, which is especially beneficial for examining languages lacking annotated datasets or benchmarks.

4.9.1 Method

For interpreting BLOOM's multilingual generalizing abilities, we utilize the "Universal Probing" framework[42] for systematic probing analysis in 104 languages and 80 morphosyntactic features (Serikov et al., 2022). The framework provides SentEval-style (Conneau et al., 2018) probing setup and datasets for each language available in Universal Dependencies (UD; Nivre et al., 2016). We consider the following 17 languages from 7 language families present in BLOOM's pretraining corpus (Section 3.1) and UD treebanks: Arabic (Afro-Asiatic), Bambara (Mande), Basque (language isolate), Bengali, Catalan, English, French, Hindi, Marathi, Portuguese, Spanish, Urdu (Indo-European), Chinese (Sino-Tibetan), Indonesian (Austronesian), Tamil (Dravidian), Wolof, Yoruba (Niger-Congo). Our setup covers 38 morphosyntactic features in total, which represent language-specific linguistic information. We provide a dataset sample in Table 11.

The probing procedure is conducted as follows. First, we compute `<s>`-pooled representations of the input sentence at each layer of the 1.7B-parameter BLOOM variant

42. `github.com/bigscience-workshop/massive-probing-framework`

Language	Label	Sentence
English	Sing	The **scheme** makes money through sponsorship and advertising .
	Plur	Still , there are **questions** left unanswered .
Spanish	Sing	Eligi**o** no ir tras un tercer período en el siguiente ciclo de elecciones .
	Plur	Todavía quedan **preguntas** sin responder .

Table 11: Examples of the Number task in English and Spanish. The subject number indicator is highlighted in **bold**. The task is to predict if the sentence includes a singular subject number (upper sentence) and a plural subject number (bottom sentence).

("BLOOM 1B7") and BLOOM (with 176B parameters). Second, we train a binary logistic regression classifier to predict a presence of a morphosyntactic feature in the sentence. Logistic regression is chosen due to its higher selectivity as opposed to non-linear probing classifiers (Hewitt and Liang, 2019). We use the original UD training, validation, and test splits here. Third, the probing performance is evaluated by F_1 weighted score due to target class imbalance for most probing tasks. The results are averaged across three runs with different random seeds.

Baselines We compare the probing performance with random guessing and logistic regression classifiers trained on the following TF-IDF features (Salton and Yang, 1973): word unigrams, character N-grams, BPE[43] token N-grams, and SentencePiece[44] (SP; Kudo and Richardson, 2018) token N-grams. We use the N-gram range $\in [1; 4]$ and limit the TF-IDF vocabularies to top-250k features.

Correlation We run statistical tests to analyze correlations between the probing performance and linguistic, dataset, and model configuration criteria:

- Language script: the results are divided into two groups by the language script – Latin and others (Devanagari, Tamil, and Arabic). Here, we use the non-parametric test Mann-Whitney U (Mann and Whitney, 1947).

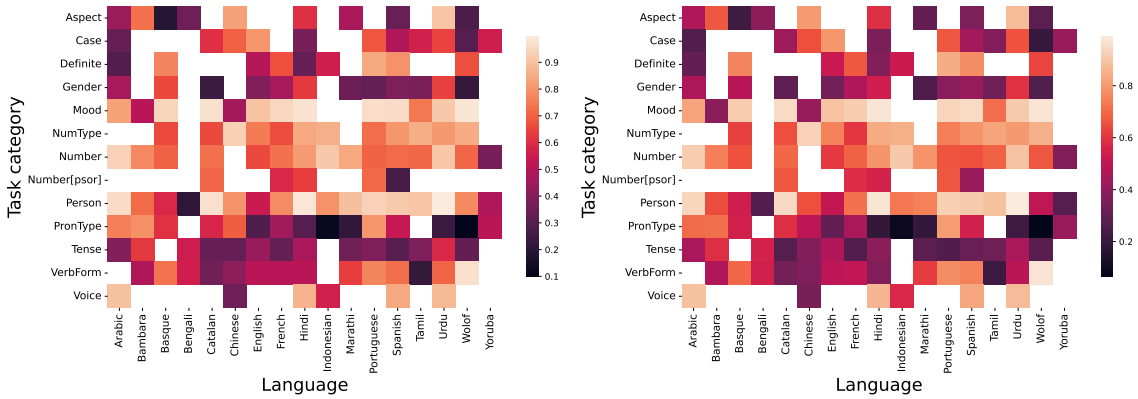
- Language family: the results are divided into 7 groups by the language family. We apply the ANOVA to analyze the variance between the groups.

- Probing and pretraining dataset size: we run the Pearson correlation coefficient test (Pearson, 1895) to compute correlation between the probing performance and these data configuration criteria.

- Effect of the model size: the results are divided into two groups by the BLOOM version. Here, we use the Mann-Whitney U test to see if there is a correlation between the number of parameters and the probing results.

	BLOOM-1B7	BLOOM	Random	TF-IDF (Char)	TF-IDF (Word)	TF-IDF (BPE)	TF-IDF (SP)
Arabic	**0.66** ±0.27	0.64 ±0.27	0.49 ±0.013	0.41 ±0.44	0.4 ±0.44	0.41 ±0.44	0.41 ±0.44
Bambara	**0.64** ±0.16	0.59 ±0.16	0.45 ±0.1	0.52 ±0.46	0.45 ±0.47	0.48 ±0.49	0.49 ±0.49
Basque	**0.68** ±0.19	0.62 ±0.19	0.49 ±0.03	0.41 ±0.43	0.44 ±0.46	0.48 ±0.44	0.41 ±0.46
Bengali	0.42 ±0.15	0.45 ±0.12	0.35 ±0.23	0.63 ±0.48	0.37 ±0.44	0.41 ±0.32	**0.76** ±0.28
Catalan	**0.65** ±0.25	0.61 ±0.26	0.34 ±0.01	0.24 ±0.38	0.24 ±0.39	0.24 ±0.39	0.24 ±0.39
Chinese	**0.66** ±0.25	0.50 ±0.21	0.55 ±0.03	0.03 ±0.05	0.11 ±0.28	0.04 ±0.06	0.03 ±0.05
English	**0.57** ±0.24	**0.57** ±0.24	0.43 ±0.03	0.45 ±0.43	0.46 ±0.43	0.45 ±0.43	0.44 ±0.44
French	**0.61** ±0.23	0.57 ±0.22	0.44 ±0.02	0.32 ±0.43	0.32 ±0.43	0.32 ±0.43	0.33 ±0.44
Hindi	**0.63** ±0.23	0.6 ±0.25	0.48 ±0.03	0.53 ±0.46	0.55 ±0.47	0.53 ±0.46	0.53 ±0.46
Indonesian	**0.65** ±0.27	0.6 ±0.27	0.48 ±0.05	0.41 ±0.46	0.43 ±0.45	0.41 ±0.46	0.45 ±0.45
Marathi	**0.57** ±0.25	0.48 ±0.24	0.32 ±0.09	0.44 ±0.47	0.46 ±0.46	0.44 ±0.47	0.44 ±0.47
Portugese	**0.67** ±0.23	0.63 ±0.26	0.4 ±0.03	0.48 ±0.48	0.49 ±0.48	0.48 ±0.48	0.48 ±0.48
Spanish	**0.66** ±0.24	0.65 ±0.24	0.42 ±0.02	0.35 ±0.42	0.35 ±0.44	0.36 ±0.44	0.36 ±0.43
Tamil	**0.57** ±0.25	0.51 ±0.27	0.43 ±0.05	0.51 ±0.44	0.53 ±0.44	0.5 ±0.44	0.5 ±0.44
Urdu	**0.75** ±0.21	0.70 ±0.24	0.43 ±0.02	0.39 ±0.48	0.39 ±0.47	0.39 ±0.48	0.39 ±0.48
Wolof	**0.51** ±0.32	0.47 ±0.32	0.41 ±0.02	0.26 ±0.39	0.25 ±0.39	0.3 ±0.43	0.27 ±0.39
Yoruba	**0.48** ±0.07	0.36 ±0.07	0.43 ±0.06	0.33 ±0.45	0.09 ±0.05	0.16 ±0.11	0.09 ±0.05

Table 12: Probing performance (F_1 averaged by layers) of the BLOOM-based classifiers and count-based baselines. The results are averaged over probing tasks, and three experiment runs within each language. Standard deviation is determined by the results along the language tasks.



(a) BLOOM-1B7 (b) BLOOM

Figure 12: Probing classifiers' results by language and task category. White squares denote that the morphosyntactic category is not represented in the language.

4.9.2 Results

Probing Table 12 presents the results of probing experiments averaged over the probing tasks and experiment runs within each language. The overall pattern is that BLOOM-1B7 performs on par or better than BLOOM, and both LLMs outperform the count-based baselines. In particular, the LLMs achieve more robust performance on Arabic, Basque, and Indo-European languages (e.g., Catalan, French, Hindi, Portuguese, Spanish, and Urdu),

43. BertTokenizer: `hf.co/bert-base-multilingual-cased`
44. XLMRobertaTokenizer: `hf.co/xlm-roberta-base`